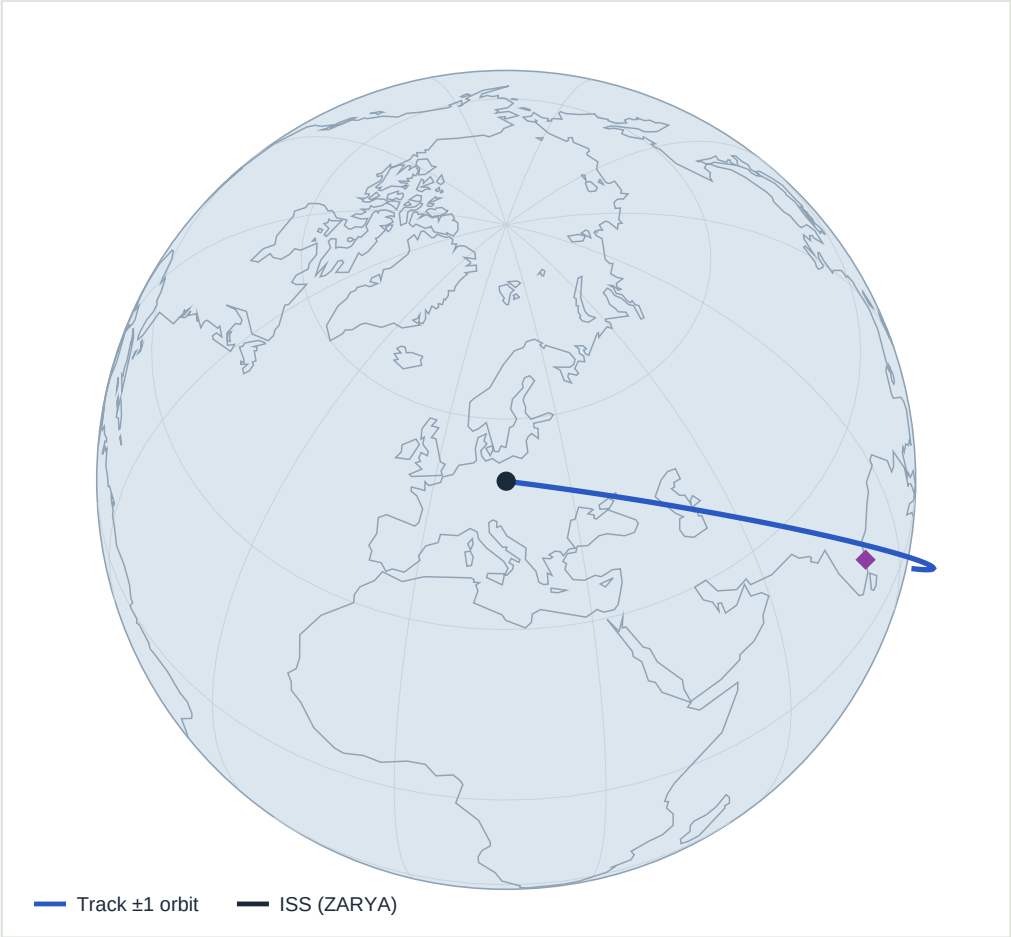
ISS (ZARYA)

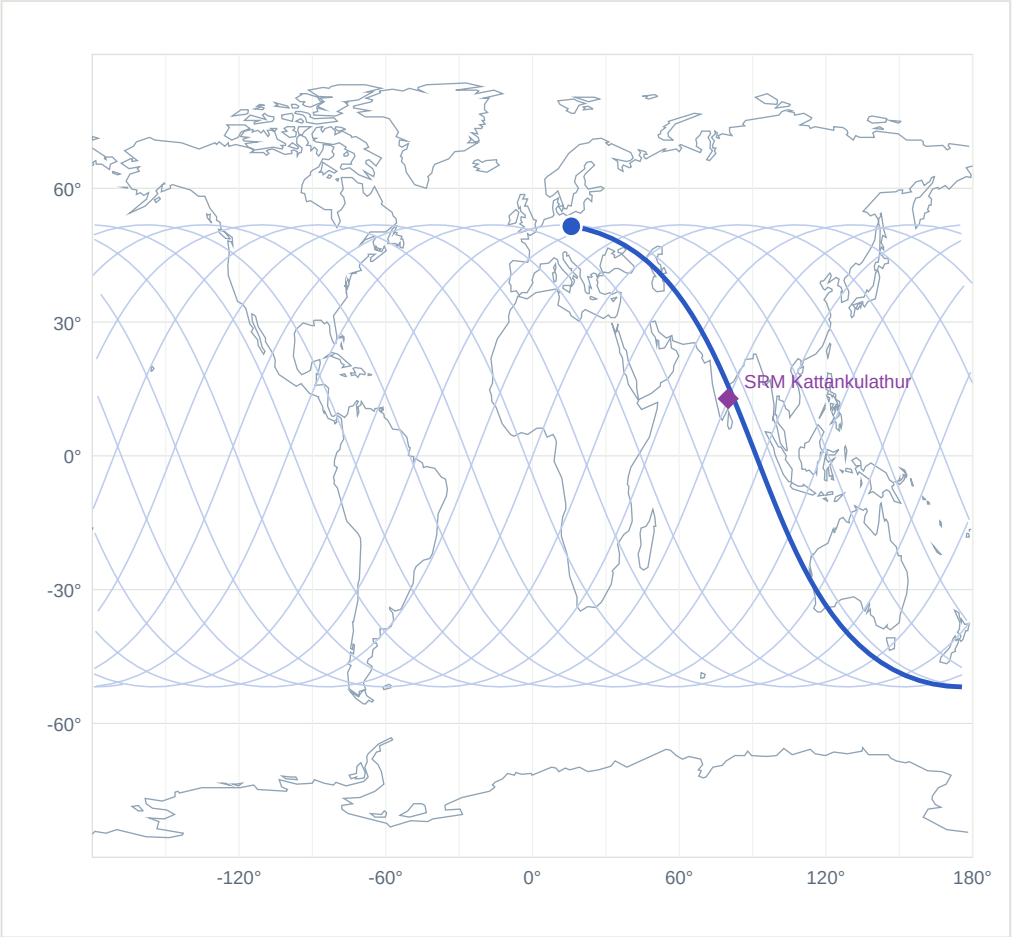
NORAD 25544 · window 14 Feb 2024 12:25 UTC + 24 h · ground station SRM Kattankulathur

Tracking | Passes | Telemetry & health | Telemetry | Fleet | Orbital elements

This TLE is 998 days old, so the view starts at its epoch (14 Feb 2024 12:25 UTC). Use live data (--norad) for current positions.

Time (UTC)	Latitude	Longitude	Altitude	Speed	Sun
12:25	51.46°	15.88°	420.8 km	7.667 km/s	Sunlit





Over this window the altitude ranges 414–440 km, the satellite is sunlit 78 % of the time across 15 eclipses (typically 20 min), and the mean solar beta angle is -63.8°.

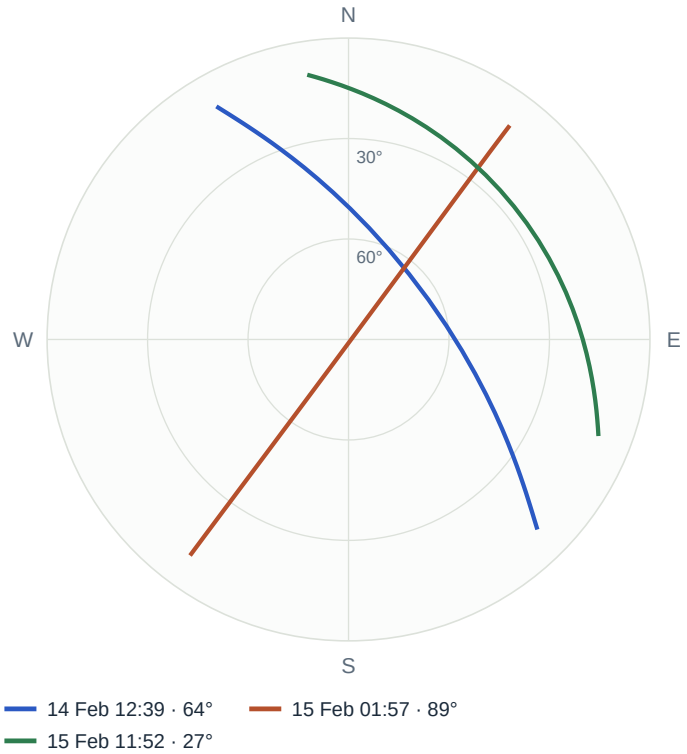
ISS (ZARYA)

NORAD 25544 · window 14 Feb 2024 12:25 UTC + 24 h · ground station SRM Kattankulathur

Passes over SRM Kattankulathur (mask 10°)

Rise (UTC)	Culmination	Set	Max elevation	Duration	Azimuth
14 Feb 12:39:34	12:42:51	12:46:08	63.7°	6.6 min	330° to 135°
15 Feb 01:57:48	02:01:09	02:04:31	89.4°	6.7 min	216° to 37°
15 Feb 11:52:11	11:55:03	11:57:55	26.6°	5.7 min	351° to 111°

Sky plot (zenith at centre)

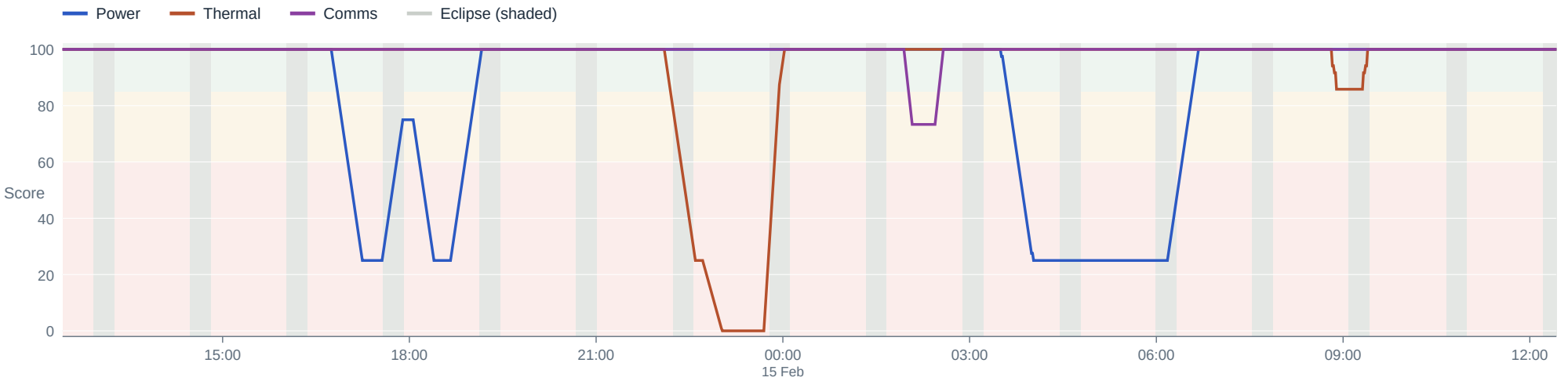


ISS (ZARYA)

NORAD 25544 · window 14 Feb 2024 12:25 UTC + 24 h · ground station SRM Kattankulathur

SIMULATED — not real spacecraft telemetry. Values are generated from the orbit geometry to demonstrate the monitoring pipeline; a TLE contains no spacecraft health data.

Overall health, end of window	Power health, end of window	Thermal health, end of window	Comms health, end of window
100 / 100	100 / 100	100 / 100	100 / 100
Lowest in window: 0 (alert)	Lowest in window: 25 (alert)	Lowest in window: 0 (alert)	Lowest in window: 73 (caution)



Limit events

Channel	Severity	Start (UTC)	End (UTC)	Min
Bus temperature	alert	14 Feb 22:43	14 Feb 23:26	44
Battery temperature	caution	14 Feb 22:45	14 Feb 22:45	1
Battery temperature	alert	14 Feb 22:47	14 Feb 23:26	40
No signal while in view	alert	15 Feb 01:57	15 Feb 02:04	8
Panel temperature	alert	15 Feb 08:49	15 Feb 08:49	1
Panel temperature	alert	15 Feb 08:53	15 Feb 08:53	1

Detector performance vs injected faults

Detector	Precision	Recall	False alarms	Faults caught
Limit checks only	1.000	0.148	0	3 of 5
Isolation Forest only	0.000	0.000	0	0 of 5
Forest + envelope guard	0.985	0.929	5	4 of 5
Limits + ML (used)	0.986	0.951	5	5 of 5

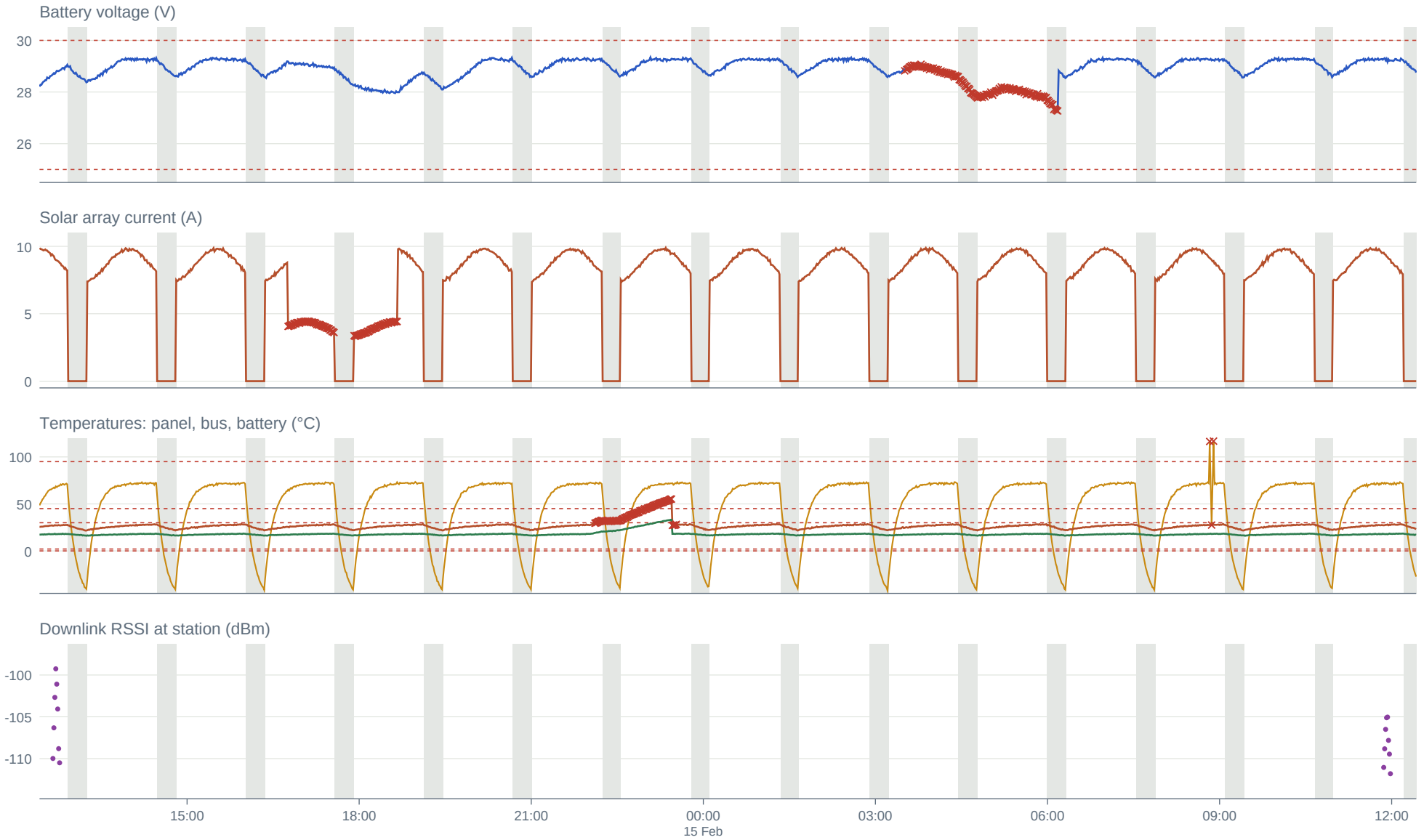
ISS (ZARYA)

NORAD 25544 · window 14 Feb 2024 12:25 UTC + 24 h · ground station SRM Kattankulathur

Tracking | Passes | Telemetry & health | **Telemetry** | Fleet | Orbital elements

Panel | Bus | Battery | Red limit | Detector flag | Eclipse

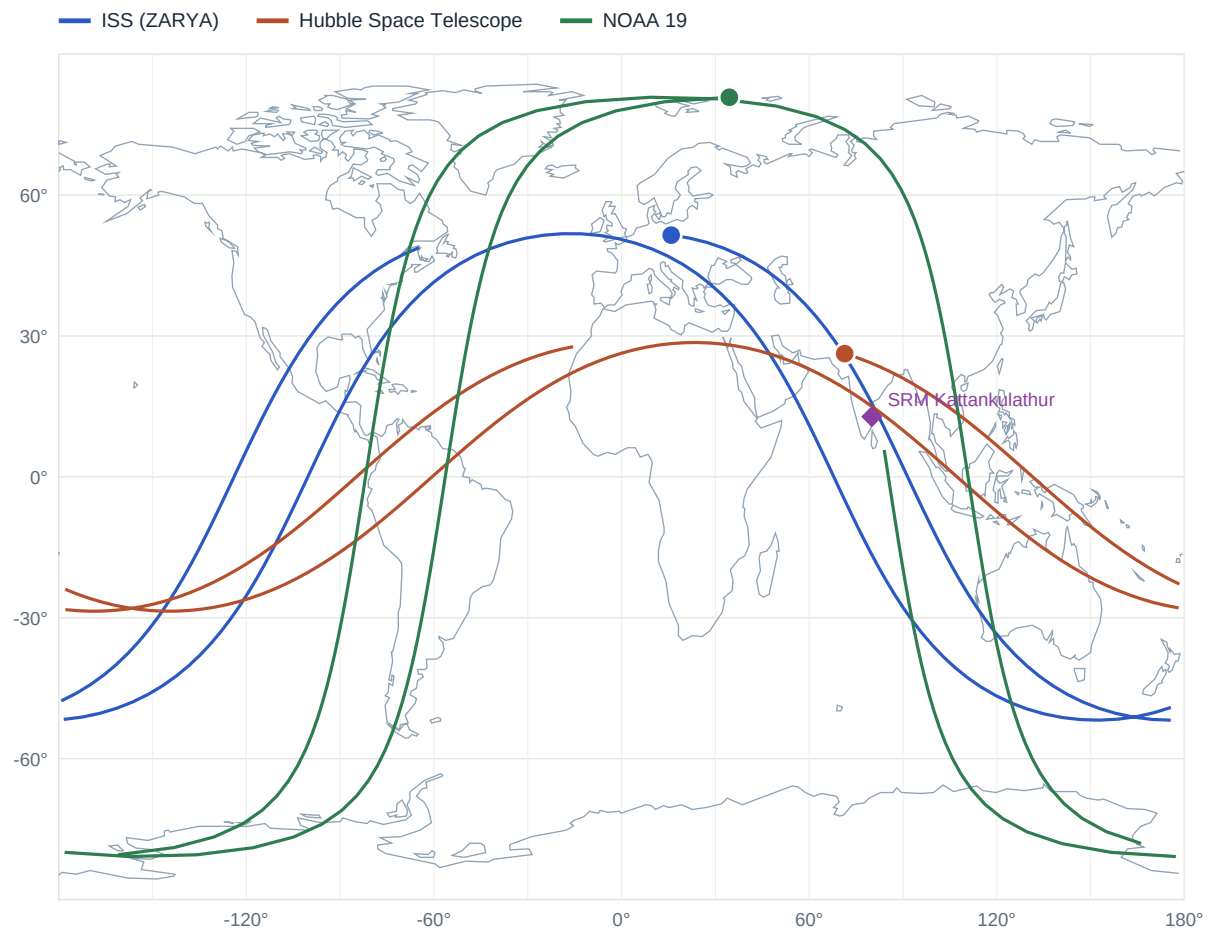
SIMULATED — not real spacecraft telemetry



ISS (ZARYA)

NORAD 25544 · window 14 Feb 2024 12:25 UTC + 24 h · ground station SRM Kattankulathur

Satellite	NORAD	Latitude	Longitude	Altitude	Inclination	Period	TLE age
ISS (ZARYA)	25544	51.46°	15.88°	420.8 km	51.64°	92.94 min	0.00 d
HUBBLE SPACE TELESCOPE	20580	26.24°	71.35°	517.3 km	28.47°	94.97 min	0.35 d
NOAA 19	33591	80.81°	34.47°	855.0 km	99.19°	101.96 min	0.02 d



All passes over SRM Kattankulathur

Satellite	Rise (UTC)	Max el	Min
Hubble	14 Feb 12:26	17°	5.3
ISS (ZARYA)	14 Feb 12:39	64°	6.6
Hubble	14 Feb 14:05	65°	8.0
NOAA 19	14 Feb 15:22	72°	10.9
Hubble	14 Feb 15:47	20°	6.1
ISS (ZARYA)	15 Feb 01:57	89°	6.7
Hubble	15 Feb 03:43	15°	4.6
NOAA 19	15 Feb 03:59	68°	10.9
Hubble	15 Feb 05:21	89°	8.0
Hubble	15 Feb 07:03	21°	6.1
ISS (ZARYA)	15 Feb 11:52	27°	5.7
Hubble	15 Feb 12:08	19°	5.9

No overlapping passes: one antenna can serve the whole fleet.

ISS (ZARYA)

NORAD 25544 · window 14 Feb 2024 12:25 UTC + 24 h · ground station SRM Kattankulathur

Element	Value
Semi-major axis	6796.66 km
Eccentricity	0.000670
Inclination	51.6416°
RAAN	247.4627°
Argument of perigee	130.5360°
Mean anomaly	325.0288°
Mean motion	15.493867 rev/day
Orbital period	92.94 min
Apogee altitude	423.07 km
Perigee altitude	413.96 km

Two-line element set

```
ISS (ZARYA)
1 25544U 98067A   24045.51782528  .00016717  00000-0  30323-3 0  9999
2 25544  51.6416 247.4627 0006703 130.5360 325.0288 15.49386731438786
```

Elements are TLE mean elements; semi-major axis from Kepler's third law, a = (mu / n^2)^(1/3), with mu = 398 600.4418 km^3/s^2.